

Yuming Fu

Curriculum vitae

Leiden Observatory
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🐙 [rudolffu](https://github.com/rudolffu)

Employment

- 01-06-2023 – **Postdoctoral Researcher**, *Leiden Observatory, Leiden University*, The Netherlands
- Research areas: Quasar and galaxy evolution, observational cosmology, data mining and machine learning in astronomy, astronomical data processing
 - Advisors: Dr. R.J. Bouwens (Leiden), and Prof. Dr. Karina Caputi (Groningen)
- Sep. 2021 – **CSST Postdoctoral Research Fellow**, *Kavli Institute for Astronomy and Astrophysics*,
May. 2023 *Peking University*, Beijing, China
- Advisor: Prof. Xue-Bing Wu

Academic service

- 01-06-2023 – **Active member**, *Galaxy/AGN Evolution Science Working Group; NIR Processing Function*, Euclid Consortium
- References: Prof. Lingyu Wang (Groningen), and Dr. Gianluca Polenta (ASI)
- 01-12-2025 – **Member**, *ESA Euclid Data Space User Group*, <https://euclid.dataspace.esa.int/eds-user-group>
- Reference: Dr. Valeria Pettorino (ESA)
- 2021– **Peer reviewer**, *A&A, ApJ, ApJS, ATI, Nat Astron, RAA*

Education

- Sep. 2016 – **Ph.D. in Astrophysics**, *Department of Astronomy, Peking University*, Beijing
- Jul. 2021
- Supervisor: Prof. Xue-Bing Wu
 - Thesis: A Survey for Quasars behind the Galactic Plane
 - Outstanding Graduate of Beijing Higher Education Institutions (2021)
- Apr. 2019 – **Visiting Ph.D. student**, *Leiden Observatory*, Leiden
- Aug. 2019
- Supervisor: Prof. Dr. Anthony G. A. Brown
- Sep. 2012 – **Bachelor of Science (Geophysics)**, *School of Geophysics and Information Technology*,
Jul. 2016 *China University of Geosciences*, Beijing
- Supervisor: Prof. Guoming Jiang
 - Thesis: High-precision Research on the Velocity Structure of Double Seismic Zone beneath Tohoku Region in Japan
 - Outstanding Graduate of Beijing Higher Education Institutions (2016)

Teaching & supervision

- 2024– **Co-supervisor**, *PhD student T. Pan*, with Prof. Rottgering and Dr. van Weeren, Leiden Observatory
- Jul. 2022 **Lecturer**, *Lesson: Decomposition of the one-dimensional spectra of quasars and galaxies*, CSST Summer School on Galaxy Sciences, Peking University
- Sep. 2018 – **Class advisor**, *The Undergraduate Class of 2018*, Department of Astronomy, School of
Jul. 2022 Physics, Peking University

- Sep. 2019 – **Co-supervisor of seven students**, *Astronomical BSc Thesis Projects*, Peking University
 Jun. 2023
- Sep. 2019 – **Teaching assistant**, *Course: Observational Experiments for Astrophysics (Optical)*,
 Jan. 2020 Peking University

High-Performance Computing & Infrastructure

- 2017 – 2023 **Computational infrastructure engineering**, *Design, configuration, and maintenance of a system with dual Intel Xeon E5-2699 v4 CPUs (44 cores, 256 GB RAM, 150 TB storage), delivered an estimated theoretical peak performance of **1.55 TFLOPS***, Peking University
- 2022 – 2025 **Research computing user**, *Experienced in using Peking University's supercomputing clusters, 'Weiming-1' (542 TFLOPS) and 'Weiming-2' (2.71 PFLOPS), for large-scale astronomical machine learning tasks*, Peking University

Grants

- 2022 **Candidate Selections for the CSST Quasar Survey with Multi-band Photometry and Slitless Spectroscopy**, *General Program of China Postdoctoral Science Foundation, No. 2022M720266, 80,000 CNY, PI*
- 2022 – 2026 **A Survey of Quasars behind the Galactic Plane**, *NSFC Key Grant, No. 12133001, 3.10 million CNY, Co-I*

First-author papers

- [5] **Euclid Collaboration: Y. Fu**, R. Bouwens, K. I. Caputi, D. Vergani, M. Scialpi, et al. "Euclid Quick Data Release (Q1): Euclid spectroscopy of quasars. 1. Identification and redshift determination of 3500 bright quasars". 2026, *A&A*, *in press (Euclid Q1 S1)*, arXiv:2512.08803. DOI: 10.1051/0004-6361/202558490.
- [4] **Yuming Fu**, Xue-Bing Wu, R. J. Bouwens, Karina I. Caputi, Yuxuan Pang, et al. "The CatSouth Quasar Candidate Catalog for the Southern Sky and a Unified All-Sky Catalog Based on Gaia DR3". 2025, *ApJS*, 279, 2, 54. DOI: 10.3847/1538-4365/ade999.
- [3] **Yuming Fu**, Xue-Bing Wu, Yifan Li, Yuxuan Pang, Ravi Joshi, et al. "CatNorth: An Improved Gaia DR3 Quasar Candidate Catalog with Pan-STARRS1 and CatWISE". 2024, *ApJS*, 271, 2, 54. DOI: 10.3847/1538-4365/ad2ae6.
- [2] **Yuming Fu**, Xue-Bing Wu, Linhua Jiang, Yanxia Zhang, Zhi-Ying Huo, et al. "Finding Quasars behind the Galactic Plane. II. Spectroscopic Identifications of 204 Quasars at $|b| < 20^\circ$ ". 2022, *ApJS*, 261, 2, 32. DOI: 10.3847/1538-4365/ac7f3e.
- [1] **Yuming Fu**, Xue-Bing Wu, Qian Yang, Anthony G. A. Brown, Xiaotong Feng, et al. "Finding Quasars behind the Galactic Plane. I. Candidate Selections with Transfer Learning". 2021, *ApJS*, 254, 1, 6. DOI: 10.3847/1538-4365/abe85e.

Second/third-author papers

- [9] Tong Pan, **Yuming Fu**, H. J. A. Rottgering, J. M. G. H. J. de Jong, M. J. Hardcastle, et al. "Probing the environments of FRI and FRII radio galaxies in LoTSS DR2 with galaxy clusters". 2026, *arXiv e-prints*, arXiv:2605.23750. DOI: 10.48550/arXiv.2605.23750.
- [8] Jin Qin, Xue-Bing Wu, **Yuming Fu**, Haojie Xu, Yuxuan Pang, et al. "Exploring the S_8 Tension: Insights from the CatNorth 1.5-Million Quasar Candidates". 2026, *ApJ*, *in press*, arXiv:2603.09457. DOI: 10.48550/arXiv.2603.09457.

- [7] Yunyi Choi, **Yuming Fu**, Myungshin Im, Xue-Bing Wu, Christopher A. Onken, et al. "AlIBRICQS: The Discovery of Luminous Quasars in the Northern Hemisphere". 2025, *ApJS*, 280, 2, 73. DOI: 10.3847/1538-4365/adf8ed.
- [6] Zi-qi He, **Yu-ming Fu**, Xue-bing Wu, and Ling-xue He. "Identifying FeLoBAL Quasars in SDSS DR7Q with the Convolutional Neural Network". 2025, *ChA&A*, 49, 4, 756–772. DOI: 10.1016/j.chinastron.2025.11.005.
- [5] Zhi-Ying Huo, **Yuming Fu**, Yang Huang, Haibo Yuan, Xue-Bing Wu, et al. "Finding Quasars behind the Galactic Plane. III. Spectroscopic Identifications of ~ 1300 New Quasars at $|b| \leq 20^\circ$ from LAMOST DR10". 2025, *ApJS*, 278, 1, 6. DOI: 10.3847/1538-4365/adba52.
- [4] T. Pan, **Y. Fu**, H. J. A. Rottgering, R. J. van Weeren, A. B. Drake, et al. "The environments of radio galaxies and quasars in LoTSS data release 2". 2025, *A&A*, 695, A69. DOI: 10.1051/0004-6361/202453154.
- [3] Yuxuan Pang, Xue-Bing Wu, **Yuming Fu**, et al. "A Pilot Study for the CSST Slitless Spectroscopic Quasar Survey Based on Mock Data". 2025, *ApJ*, 980, 223. DOI: 10.3847/1538-4357/adabdc.
- [2] Yuxuan Pang, Xue-Bing Wu, **Yuming Fu**, Rui Zhu, Tao Ji, et al. "Quasar identifications from the slitless spectra: a test from 3D-HST". 2025, *MNRAS*, 540, 3, 2216–2237. DOI: 10.1093/mnras/staf849.
- [1] Jun-Jie Jin, Xue-Bing Wu, **Yuming Fu**, et al. "The Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: Quasar Properties from Data Releases 6 to 9". 2023, *ApJS*, 265, 1, 25. DOI: 10.3847/1538-4365/acaf89.

Contributed Euclid papers (selected)

- [8] Euclid Collaboration, A. A. Tumborang, K. I. Caputi, P. Rinaldi, L. Bisigello, et al. "Euclid: Scaled-up little red dots and other sources with v-shaped spectral energy distributions at $z > 4$ ". 2026, *arXiv e-prints*, arXiv:2604.16178.
- [7] Euclid Collaboration, T. Matamoro Zatarain, S. Fotopoulou, F. Ricci, et al. "Euclid Quick Data Release (Q1). The active galaxies of Euclid". 2025, *A&A*, *in press (Euclid Q1 SI)*, arXiv:2503.15320. DOI: 10.1051/0004-6361/202554619.
- [6] Euclid Collaboration, Y. Mellier, Abdurro'uf, J.A. Acevedo Barroso, et al. "Euclid - I. Overview of the Euclid mission". 2025, *A&A*, 697, A1. DOI: 10.1051/0004-6361/202450810.
- [5] Euclid Collaboration, R. Navarro-Carrera, and et al. "Euclid Quick Data Release (Q1): Identification of massive galaxy candidates at the end of the Epoch of Reionisation". 2025, *A&A*, *submitted*, arXiv:2511.11943. DOI: 10.48550/arXiv.2511.11943.
- [4] Euclid Collaboration, G. Polenta, M. Frailis, A. Alavi, et al. "Euclid Quick Data Release (Q1). NIR processing and data products". 2025, *A&A*, *in press (Euclid Q1 SI)*, arXiv:2503.15304. DOI: 10.1051/0004-6361/202554657.
- [3] Euclid Collaboration, M. Siudek, M. Huertas-Company, M. Smith, et al. "Euclid Quick Data Release (Q1) Exploring galaxy properties with a multi-modal foundation model". 2025, *A&A*, *in press (Euclid Q1 SI)*, arXiv:2503.15312. DOI: 10.1051/0004-6361/202554611.

- [2] Euclid Collaboration, F. Tarsitano, S. Fotopoulou, M. Banerji, et al. “Euclid Quick Data Release (Q1) First study of red quasars selection”. 2025, *A&A*, *in press (Euclid Q1 S1)*, arXiv:2503.15319. DOI: 10.1051/0004-6361/202554591.
- [1] L. Ulivi and et al. “Euclid: A machine-learning search for dual and lensed AGN at sub-arcsec separations”. 2025, *A&A*, *submitted*, arXiv:2508.19494. DOI: 10.48550/arXiv.2508.19494.

Other co-authored papers

- [18] Bing Lyu, Xue-Bing Wu, Jun-Jie Jin, **Yuming Fu**, Yuxuan Pang, et al. “The Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: Quasar Properties from Data Releases 10 to 12”. 2026, *ApJS*, 282, 2, 72. DOI: 10.3847/1538-4365/ae2b6e.
- [17] Jing-Hang Shi, Yanxia Zhang, Changhua Li, Jingyi Zhang, Zihan Kang, et al. “Unveiling Quasars in Gaia DR3: Multimodal Classification and Redshift Estimation of BP/RP Sources with Deep Learning”. 2026, *ApJS*, 283, 1, 38. DOI: 10.3847/1538-4365/ae4003.
- [16] Huimei Wang, Xue-Bing Wu, Nanyu Yao, Bing Lyu, Yuxuan Pang, et al. “Systematic Analysis of Changing-Look Active Galactic Nucleus Variability Using ZTF Light Curves”. 2026, *ApJ*, 997, 1, 100. DOI: 10.3847/1538-4357/ae1fdc.
- [15] Shouyi Wang, Fan Zou, Chang-Hao Chen, W. N. Brandt, Elena Gallo, et al. “A Rare Eddington-limited, Heavily Obscured Low-mass Active Galactic Nucleus Likely Triggered by a Galaxy Merger”. 2026, *ApJ*, 1003, 2, 172. DOI: 10.3847/1538-4357/ae6779.
- [14] Xu Zhang, Yanli Ai, Yanxia Zhang, Yuming Fu, Xue-Bing Wu, et al. “Finding Quasars behind the Galactic Plane. IV. Candidate Selection from Chandra with Random Forest”. 2026, *ApJ*, 1002, 1, 76. DOI: 10.3847/1538-4357/ae5c03.
- [13] Rui Zhu, Xue-Bing Wu, Yuxuan Pang, and **Yuming Fu**. “QHSC: The Quasar Candidate Catalog for the Hyper Suprime-Cam Subaru Strategic Program”. 2026, *ApJS*, 282, 2, 38. DOI: 10.3847/1538-4365/ae2099.
- [12] Bing Lyu et al. “The changing-look AGN SDSS J101152.98+544206.4 is returning to a type I state”. 2025, *A&A*, 693, A173. DOI: 10.1051/0004-6361/202451699.
- [11] Heng Wang, Yanli Ai, Yanxia Zhang, **Yuming Fu**, Wenfeng Wen, et al. “Highly Variable Quasar Candidates Selected from 4XMM-DR13 with Machine Learning”. 2025, *ApJ*, 985, 1, 23. DOI: 10.3847/1538-4357/adc7b8.
- [10] Di Wu, Zizhao He, Nan Li, Shenzhe Cui, Yuming Fu, et al. “CURLING – III. Identifying Candidates of Wide-separation Gravitationally Lensed Quasars from the CatNorth Catalogue”. 2025, *arXiv e-prints*, arXiv:2509.17071. DOI: 10.48550/arXiv.2509.17071.
- [9] Xu Zhang, Yanli Ai, Yanxia Zhang, **Yuming Fu**, Xue-Bing Wu, et al. “Finding Quasars behind the Galactic Plane. IV. Candidate Selection from Chandra with Random Forest”. 2025, *arXiv e-prints*, arXiv:2512.23060. DOI: 10.48550/arXiv.2512.23060.
- [8] Chengqi Liu et al. “Forecasting supernova observations with the CSST: I. Photometric samples”. 2024, *Science China Physics, Mechanics, and Astronomy*, 67, 11, 119512. DOI: 10.1007/s11433-024-2456-x.
- [7] Siqi Liu, A. -Li Luo, Zhenya Zheng, Wei Zhang, **Yu-Ming Fu**, et al. “The origin of the X-ray luminosity of the green pea galaxies: X-ray binaries or active galactic nuclei?” 2024, *A&A*, 689, A170. DOI: 10.1051/0004-6361/202449406.

- [6] Qinchun Ma, Yuhan Wen, Xue-Bing Wu, Huapeng Gu, and **Yuming Fu**. “H α Time Delays of Active Galactic Nuclei from the Zwicky Transient Facility Broadband Photometry”. 2024, *ApJ*, 966, 1, 5. DOI: 10.3847/1538-4357/ad34d6.
- [5] Qinchun Ma, Xue-Bing Wu, Huapeng Gu, Yuhan Wen, and **Yuming Fu**. “The H α Broad-band Photometric Reverberation Mapping of Four Seyfert 1 Galaxies”. 2023, *ApJ*, 949, 1, 22. DOI: 10.3847/1538-4357/acc4c1.
- [4] Jiang-Tao Li, Feige Wang, Jinyi Yang, Yuchen Zhang, **Yuming Fu**, et al. “Chandra Detection of Three X-Ray Bright Quasars at $z > 5$ ”. 2021, *ApJ*, 906, 2, 135. DOI: 10.3847/1538-4357/abc750.
- [3] Su Yao et al. “The Large Sky Area Multi-object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: The Fourth and Fifth Data Releases”. 2019, *ApJS*, 240, 1, 6. DOI: 10.3847/1538-4365/aaef88.
- [2] Qian Yang et al. “Discovery of 21 New Changing-look AGNs in the Northern Sky”. 2018, *ApJ*, 862, 2, 109. DOI: 10.3847/1538-4357/aaca3a.
- [1] Qian Yang et al. “Quasar Photometric Redshifts and Candidate Selection: A New Algorithm Based on Optical and Mid-infrared Photometric Data”. 2017, *AJ*, 154, 6, 269. DOI: 10.3847/1538-3881/aa943c.

Softwares

- [3] **Yuming Fu**. *specbox: a simple tool to manipulate and visualize UV/optical/NIR spectra for astronomical research*. Version v1.0.0. Feb. 2026. DOI: 10.5281/zenodo.18642758.
- [2] **Yuming Fu**. *QSOFITMORE: a python package for fitting UV-optical spectra of quasars*. Version v1.2.2. June 2025. DOI: 10.5281/zenodo.15571037.
- [1] **Yuming Fu**. *PyFOSC: a pipeline toolbox for BFOSC/YFOSC long-slit spectroscopy data reduction*. Version v1.1.0. Apr. 2024. DOI: 10.5281/zenodo.10967240.

Conference Talks

- Dec. 2025 **Identification of 3500 Bright QSOs with Euclid Q1 Spectroscopy (invited)**, *Working with Euclid Q1: From Data Access to Science*, Madrid, Spain
- Mar. 2025 **4000 Bright Quasar Candidates from Gaia DR3 in Euclid Q1**, *ESLAB# 56 and 2025 Euclid Consortium Meeting*, Leiden, Netherlands
- Jan. 2025 **Highly Complete Bright Quasar Sample from Gaia DR3 in Euclid Q1: Insights from the CatNorth+CatSouth Catalogues**, *EUCLID – From Q1 to DR1. Joint Workshop of Euclid Local Universe, Galaxy and AGN Evolution, and Primeval Universe Science Working Groups*, Tenerife, Spain
- Jun. 2024 **Identifying the Rare FeLoBAL Quasars: Generative and Active Learning with Limited Data**, *2024 Euclid Consortium Meeting*, Rome, Italy
- Feb. 2024 **Synergies Between Euclid and CSST in Galaxy and AGN Surveys: Indications from the First Euclid Data**, *2024 Euclid Galaxy and AGN Evolution Science Working Group Workshop*, Bologna, Italy
- Dec. 2021 **Finding Iron Low-ionization Broad Absorption Line Quasars with SDSS**, *2020 Annual Meeting of Chinese Astronomical Society*, Nanchong, China
- May. 2021 **A Survey for Quasars behind the Galactic Plane**, *The 23rd CAS Guoshoujing Symposium on Galaxies and Cosmology*, Hangzhou, China
The Best Oral Presentation

Apr. 2019 **Uncovering Hidden Quasars behind the Galactic Plane with AllWISE, Pan-STARRS and Gaia**, *ESLAB# 53: The Gaia Universe*, Noordwijk, Netherlands

Observations

- 2017 – 2022 **The 2.16-meter Telescope**, *Xinglong Observatory, National Astronomical Observatories, Chinese Academy of Sciences (NAOC)*, PI/Co-I, on-site/remote
Allocated time: > 350 hours
- 2017 – 2022 **The 2.4-meter Telescope**, *Lijiang Observatory, Yunnan Observatories (YNAO), Chinese Academy of Sciences*, PI/Co-I, on-site/remote
Allocated time: > 180 hours
- 2019 – 2021 **The McGraw-Hill 1.3-m Telescope**, *MDM Observatory*, PI/Co-I, remote
Allocated time: > 100 hours
- 2018 – 2024 **The 200-inch Hale Telescope (P200)**, *Palomar Observatory*, PI/Co-I, remote
Allocated time: > 90 hours

Training/Schools

- Jul. 2018 **The 2nd East Asian Workshop on Astrostatistics**, *Purple Mountain Observatory*, Nanjing, China
- Nov. 2017 **The Ninth Xinglong Observational Astrophysics Workshop: High Resolution Spectroscopy**, *Xinglong Observatory, NAOC, Xinglong, China*
- Sep. 2017 **TIARA Summer School on Astrostatistics and Big Data**, *TIARA, ASIAA, Taipei, China*
- Jul. 2017 **LAMOST Users Workshop 2017**, *NAOC/UCAS, Beijing, China*
- Aug. 2016 **The Sixth Xinglong Observational Astrophysics Workshop: Time Domain Astronomy**, *Xinglong Observatory, NAOC, Xinglong, China*

Skills

- Observation **Photometric and spectroscopic observations. Data reduction with IRAF, Python, etc.**
- Coding **Python, MATLAB, Shell, Fortran, C**
- Database **ADQL, TOPCAT, MySQL, ClickHouse**
- Big Data **Data Mining, Statistical Machine Learning, Deep Learning, LLM**

Outreach

- Jan. 2024 **Public Talk: The Beauty of the Space II: the Universe as seen by JWST and Euclid**, *Library of Jianyang City, Chengdu*
- Mar. 2022 **Public Talk: Quasars in the Distant Universe**, *AstroSalon# 6 of the Peking University Youth Astronomy Society, Beijing*
- Jan. 2022 **Public Talk: The Beauty of the Space**, *Library of Jianyang City, Chengdu*
- Jan. 2020 **Public Talk: A Brief History of Black Holes**, *Library of Jianyang City, Chengdu*
- Jan. 2019 **Public Talk: The Big Eyes: A Glimpse at the Telescopes**, *Library of Jianyang City, Chengdu*
- Feb. 2018 **Public Talk: Astronomy in Everyday Life**, *Hangzhou Arts and Crafts Museum, Hangzhou*

Honors & Awards

2020 **Merit Student**, *Peking University*, Beijing
2018, 2020 **Dedicated Scholarship for Outstanding PhD Students**, *Peking University*, Beijing
2020 **Outstanding Teaching Assistant**, *Peking University*, Beijing
2017, 2019 **Award for Scientific Research**, *Peking University*, Beijing
2017 **The Third Prize of CASC Scholarship**, *Peking University*, Beijing